Development Setup

This guide covers setting up the development environment, running the application, and preparing the database.

Prerequisites Check

```
# Check Node.js (v20+ required)
node --version

# Check npm (v9+ required)
npm --version

# Check PostgreSQL (v14+ required)
psql --version
```

Step 1: Prepare the Database

Start PostgreSQL and create a database:

```
# Connect to PostgreSQL
psql -U postgres
```

```
# Create database
CREATE DATABASE kuccps_ticketing;
\q
```

Step 2: Configure Environment Variables

Follow the Configuration guide to set up:

- `backend/.env` — database URL, JWT secret, API key
- `frontend/.env` — API base URL

Step 3: Generate Prisma Client

```
cd backend
npx prisma generate
```

This generates the Prisma Client based on `schema.prisma`.

Step 4: Run Database Migrations

```
cd backend
npx prisma migrate dev
```

For production:

```
cd backend
npx prisma migrate deploy
```

Step 5: Run Preflight Checks

```
cd backend
npm run preflight
```

The preflight script validates the environment configuration and database connectivity.

Step 6: Seed the Database

```
cd backend
npx prisma db seed
```

This creates:

Entity	Details
Department	ICT department
Admin user	`admin` / `admin123`
Staff user	`itstaff` / `staff123`

Entity	Details
Default settings	Organization name, colors, SMTP defaults
Email templates	Ticket lifecycle templates

> **Warning:** Change default passwords before using the system beyond local testing.

Step 7: Start the Backend

```
cd backend
npm start
```

The API starts on `http://localhost:5000`.

Verify it is running:

```
curl http://localhost:5000/health
```

Expected response:

```
{
  "status": "OK",
  "message": "KUCCPS Ticketing System API is running"
}
```

Step 8: Start the Frontend

In a new terminal:

```
cd frontend
npm run dev
```

The frontend starts on `http://localhost:5173` or `http://localhost:5174`.

Step 9: Access the Application

Open `http://localhost:5173` in your browser and log in with:

- **Username:** `admin`
- **Password:** `admin123`

Available Scripts

Command	Directory	Purpose
`npm start`	`backend/`	Start Express server
`npm run dev`	`frontend/`	Start Vite dev server
`npm run build`	`frontend/`	Build for production

Command	Directory	Purpose
`npm run preview`	`frontend/`	Preview production build
`npm run lint`	`frontend/`	Run ESLint
`npx prisma studio`	`backend/`	Open Prisma database browser
`npx prisma migrate dev`	`backend/`	Create and apply migrations
`npx prisma db seed`	`backend/`	Seed database with default data

Hot Reloading

- **Backend:** Changes to `server.js` and route files require a server restart.
- **Frontend:** Vite provides instant hot module replacement (HMR) for React components.

Database Browser

Use Prisma Studio to view and edit data:

```
cd backend
npx prisma studio
```

This opens a web UI at `http://localhost:5555`.

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